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Key Challenges in Drone Evolution

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7.1 Introduction

The evolution of drone technology from its early experimental forms to the sophisticated fifth-generation unmanned aerial systems (UAS) has been driven as much by innovation as by the challenges encountered along the way. Each generation of drones has faced distinct technological, regulatory, and societal barriers that have collectively shaped the pace, direction, and purpose of development. Understanding these challenges provides critical insight into how drones have transformed from simple remotely piloted aircraft to intelligent, autonomous aerial systems capable of complex, multi-domain operations.

7.1.1 Technological constraints in drone advancement

From the beginning, drone evolution has been constrained by technological limitations, including propulsion efficiency, flight stability, endurance, and communication reliability. Early generations struggled with power sources, limited range, and mechanical precision. As automation advanced, AI integration, data processing speed, cybersecurity, and energy management became central challenges, particularly in fifth-generation drones.

Parallel to these, regulatory challenges have persisted across all eras. The absence of standardized frameworks for airspace management, privacy